

27 Which wavelength of electromagnetic radiation in free space could be green light?

- A $4.9 \times 10^{-9} \text{ m}$ B $6.6 \times 10^{-9} \text{ m}$ C $5.5 \times 10^{-7} \text{ m}$ D $8.6 \times 10^{-7} \text{ m}$

28 Which group contains only waves that can be polarised?

- A infrared waves, radio waves, sound waves
B visible light waves, microwaves, radio waves
C visible light waves, radio waves, sound waves
D microwaves, visible light waves, sound waves

29 One wave has an amplitude of $2A$. A second wave has an amplitude of $\frac{A}{2}$. The waves are otherwise identical.

The two waves travel in opposite directions and overlap.

What is the ratio $\frac{\text{maximum amplitude of combined wave}}{\text{minimum amplitude of combined wave}}$?

- A $\frac{3}{2}$ B $\frac{5}{3}$ C $\frac{5}{2}$ D 4

30 Light of wavelength 680 nm is incident normally on a diffraction grating with $450 \text{ lines mm}^{-1}$.

What is the angle of diffraction of the second-order maximum in the diffraction pattern that is produced?

- A 1.8° B 3.5° C 18° D 38°

31 A hollow tube is closed at one end and open at the other.

A stationary sound wave of the lowest possible frequency, 820 Hz , is produced in the tube.

The speed of sound in air is 330 ms^{-1} .

What is the length of the tube?

- A 10 cm B 20 cm C 40 cm D 160 cm

32 Which expression gives the definition of resistance?

- A current divided by potential difference
B current multiplied by potential difference
C potential difference divided by current
D resistivity multiplied by length